Ana Laura Pereira 6° CD

1. Atualização do Sistema

```
@AnaLaura1997 ➜/workspaces/infraestrutura_big_data (main) $ sudo apt update
Hit:1 https://packages.microsoft.com/repos/microsoft-ubuntu-noble-prod noble InRelease
Get:2 https://dl.yarnpkg.com/debian stable InRelease
Hit:3 https://repo.anaconda.com/pkgs/misc/debrepo/conda stable InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:7 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Err:2 https://dl.yarnpkg.com/debian stable InRelease
  The following signatures couldn't be verified because the public key is not available: NO_PUBKE
Y 62D54FD4003F6525
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1895 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1260 kB]
Reading package lists... Done
W: GPG error: https://dl.yarnpkg.com/debian stable InRelease: The following signatures couldn't b
e verified because the public key is not available: NO_PUBKEY 62D54FD4003F6525
E: The repository 'https://dl.yarnpkg.com/debian stable InRelease' is not signed.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
@AnaLaura1997 ➜/workspaces/infraestrutura_big_data (main) $ sudo apt upgrade -y
```

sudo apt update: atualiza a lista de pacotes disponíveis nos repositórios para que o sistema saiba quais versões mais recentes existem para instalar ou atualizar.

```
@AnaLaura1997 ➜/workspaces/infraestrutura_big_data (main) $ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages will be upgraded:
  libnss3 libnss3-dev libpq-dev libpq5
4 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
Need to get 1996 kB of archives.
After this operation, 5120 B of additional disk space will be used.
Get:1 http://security.ubuntu.com/ubuntu noble-security/main amd64 libnss3-dev amd64 2:3.98-1ubunt
u0.1 [253 kB]
Get:2 http://security.ubuntu.com/ubuntu noble-security/main amd64 libnss3 amd64 2:3.98-1ubuntu0.1
 [1445 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security/main amd64 libpq-dev amd64 16.13-0ubuntu0.
24.04.1 [152 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 libpq5 amd64 16.13-0ubuntu0.24.
04.1 [146 kB]
Fetched 1996 kB in 0s (7208 kB/s)
debconf: unable to initialize frontend: Dialog
debconf: (Dialog frontend requires a screen at least 13 lines tall and 31 columns wide.)
debconf: falling back to frontend: Readline
(Reading database ... 58647 files and directories currently installed.)
Preparing to unpack .../libnss3-dev_2%3a3.98-1ubuntu0.1_amd64.deb ...
Unpacking libnss3-dev:amd64 (2:3.98-1ubuntu0.1) over (2:3.98-1build1) ...
Preparing to unpack .../libnss3_2%3a3.98-1ubuntu0.1_amd64.deb ...
Unpacking libnss3:amd64 (2:3.98-1ubuntu0.1) over (2:3.98-1build1) ...
```

sudo apt upgrade -y: instala automaticamente as atualizações disponíveis dos programas já instalados no sistema, sem pedir confirmação.

```
● @AnaLaura1997 ➜/workspaces/infraestrutura_big_data (main) $ pwd
  /workspaces/infraestrutura_big_data
● @AnaLaura1997 ➜/workspaces/infraestrutura_big_data (main) $ ls
  README.md
● @AnaLaura1997 ➜/workspaces/infraestrutura_big_data (main) $ ls -la
  total 16
  drwxrwxrwx+ 3 codespace root 4096 Mar  4 17:43 .
  drwxr-xrwx+ 5 codespace root 4096 Mar  4 17:43 ..
  drwxrwxrwx+ 8 codespace root 4096 Mar  4 17:43 .git
  -rw-rw-rw-  1 codespace root   25 Mar  4 17:43 README.md
```

```
● @AnaLaura1997 ➜/workspaces/infraestrutura_big_data (main) $ cd ~
● @AnaLaura1997 ➜~ $ mkdir aula_bigdata
● @AnaLaura1997 ➜~ $ cd aula_bigdata
○ @AnaLaura1997 ➜~/aula_bigdata $
```

2. Navegação no Sistema

Pwd: mostra o caminho completo do diretório onde você está no momento**.**

Ls: lista os arquivos e pastas que estão no diretório atual.

ls -la: lista todos os arquivos e pastas do diretório atual, incluindo os ocultos, mostrando também detalhes como permissões, dono, tamanho e data de modificação.

cd~: leva direto para o seu diretório pessoal

mkdir aula_bigdata: cria uma nova pasta chamada aula_bigdata no diretório

cd aula_bigdata: entra na pasta chamada aula_bigdata.

```
● @AnaLaura1997 ➜~/aula_bigdata $ touch arquivo1.txt
● @AnaLaura1997 ➜~/aula_bigdata $ echo "infraestrutura para Big Data" > arquivo1.txt
● @AnaLaura1997 ➜~/aula_bigdata $ cat arquivo1.txt
  infraestrutura para Big Data
● @AnaLaura1997 ➜~/aula_bigdata $ cp arquivo1.txt copia.txt
● @AnaLaura1997 ➜~/aula_bigdata $ mv copia.txt arquivo2.txt
● @AnaLaura1997 ➜~/aula_bigdata $ rm arquivo2.txt
○ @AnaLaura1997 ➜~/aula_bigdata $
```

3. Manipulação de Arquivos

touch arquivo1.txt : cria um arquivo vazio chamado arquivo1.txt no diretório atual

echo "Infraestrutura para Big Data" > arquivo1.txt: grava o texto dentro do arquivo arquivo1.txt, substituindo todo o conteúdo que já existia nele.

cat arquivo1.txt :mostra na tela o conteúdo que está dentro do arquivo arquivo1.txt.

cp arquivo1.txt copia.txt: cria uma cópia do arquivo arquivo1.txt com o nome copia.txt.

mv copia.txt arquivo2.txt : renomeia o arquivo copia.txt para arquivo2.txt.

rm arquivo2.txt: apaga permanentemente o arquivo arquivo2.txt do diretório atual.

4. Visualização de Processos

```
@AnaLaura1997 →~/aula_bigdata $ top
top - 17:58:02 up  1:52,  0 user,  load average: 0.92, 0.91, 1.04
Tasks:  18 total,   1 running,  17 sleeping,   0 stopped,   0 zombie
%Cpu(s):  5.8 us,  7.4 sy, 10.0 ni, 74.0 id,  2.6 wa,  0.0 hi,  0.2 si,  0.0 st
MiB Mem :   7944.6 total,    155.2 free,   2862.0 used,   5335.2 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.   5082.5 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   1992 codespa+  20   0   42.1g 369716  58752 S   1.7   4.5   0:19.61 node
   2603 codespa+  20   0 1067228  74408  48256 S   1.0   0.9   0:01.99 node
      1 codespa+  20   0    1136      0      0 S   0.0   0.0   0:00.06 docker-init
      7 codespa+  20   0    2800   1664   1664 S   0.0   0.0   0:00.20 sh
    236 root      20   0   12016   2580   1536 S   0.0   0.0   0:00.00 sshd
    340 root      20   0 1895732  26168   5248 S   0.0   0.3   0:00.44 dockerd
    517 root      20   0 1802720  21992   7936 S   0.0   0.3   0:00.15 containerd
   1244 codespa+  20   0    2800   1664   1664 S   0.0   0.0   0:00.01 sh
   1272 root      20   0    2800   1792   1792 S   0.0   0.0   0:00.00 sh
   1953 codespa+  20   0    2808   1664   1664 S   0.0   0.0   0:00.00 sh
   1963 codespa+  20   0   11.3g 114048  52608 S   0.0   1.4   0:05.62 node
   2016 codespa+  20   0 1262968  59560  47744 S   0.0   0.7   0:00.25 node
   2615 codespa+  20   0   16716  12800   3584 S   0.0   0.2   0:00.21 bash
   3052 codespa+  20   0 1183640  64980  48512 S   0.0   0.8   0:00.23 node
   3147 codespa+  20   0    2800   1664   1664 S   0.0   0.0   0:00.00 sh
   3175 root      20   0    2800   1664   1664 S   0.0   0.0   0:00.01 sh
  10598 codespa+  20   0   12348   5504   3456 R   0.0   0.1   0:00.00 top
  10635 codespa+  20   0    6112   1792   1792 S   0.0   0.0   0:00.00 sleep
```

Top: mostra em tempo real os processos que estão rodando no sistema, com uso de CPU, memória e outras informações.

- Dificuldade: Não sabia que precisava pausar

```
  0[||||||||||||                          24.8%] Tasks: 21, 70 thr, 0 kthr; 1 running
  1[||||||||||||||||                      32.7%] Load average: 0.28 0.64 0.89
Mem[||||||||||||||||||||||||||||||||||2.50G/7.76G] Uptime: 01:57:54
Swp[                                      0K/0K]

[Main] [I/O]
   PID USER      PRI  NI  VIRT   RES   SHR S  CPU%▽MEM%   TIME+  Command
  2022 codespace  20   0 1233M 59560 47744 S   0.0  0.7  0:00.00 /vscode/bin/linux-x64/072586267e68ece9a
  2023 codespace  20   0 1233M 59560 47744 S   0.0  0.7  0:00.00 /vscode/bin/linux-x64/072586267e68ece9a
  2032 codespace  20   0 1233M 59560 47744 S   0.0  0.7  0:00.00 /vscode/bin/linux-x64/072586267e68ece9a
  2054 codespace  20   0 1233M 59560 47744 S   0.0  0.7  0:00.00 /vscode/bin/linux-x64/072586267e68ece9a
  2055 codespace  20   0 1233M 59560 47744 S   0.0  0.7  0:00.00 /vscode/bin/linux-x64/072586267e68ece9a
  2056 codespace  20   0 1233M 59560 47744 S   0.0  0.7  0:00.00 /vscode/bin/linux-x64/072586267e68ece9a
  2057 codespace  20   0 1233M 59560 47744 S   0.0  0.7  0:00.00 /vscode/bin/linux-x64/072586267e68ece9a
  2175 codespace  20   0 1233M 59560 47744 S   0.0  0.7  0:00.00 /vscode/bin/linux-x64/072586267e68ece9a
  2176 codespace  20   0 1233M 59560 47744 S   0.0  0.7  0:00.04 /vscode/bin/linux-x64/072586267e68ece9a
  2603 codespace  20   0 1043M 75324 48256 S   0.0  0.9  0:02.13 /vscode/bin/linux-x64/072586267e68ece9a
  2604 codespace  20   0 1043M 75324 48256 S   0.0  0.9  0:00.00 /vscode/bin/linux-x64/072586267e68ece9a
  2605 codespace  20   0 1043M 75324 48256 S   0.0  0.9  0:00.08 /vscode/bin/linux-x64/072586267e68ece9a
  2606 codespace  20   0 1043M 75324 48256 S   0.0  0.9  0:00.10 /vscode/bin/linux-x64/072586267e68ece9a
  2607 codespace  20   0 1043M 75324 48256 S   0.0  0.9  0:00.11 /vscode/bin/linux-x64/072586267e68ece9a
  2608 codespace  20   0 1043M 75324 48256 S   0.0  0.9  0:00.09 /vscode/bin/linux-x64/072586267e68ece9a
  2609 codespace  20   0 1043M 75324 48256 S   0.0  0.9  0:00.00 /vscode/bin/linux-x64/072586267e68ece9a
  2610 codespace  20   0 1043M 75324 48256 S   0.0  0.9  0:00.00 /vscode/bin/linux-x64/072586267e68ece9a
  2611 codespace  20   0 1043M 75324 48256 S   0.0  0.9  0:00.00 /vscode/bin/linux-x64/072586267e68ece9a
  2612 codespace  20   0 1043M 75324 48256 S   0.0  0.9  0:00.00 /vscode/bin/linux-x64/072586267e68ece9a
  2613 codespace  20   0 1043M 75324 48256 S   0.0  0.9  0:00.00 /vscode/bin/linux-x64/072586267e68ece9a
  2615 codespace  20   0 16716 12800  3584 S   0.0  0.2  0:00.23 /bin/bash --init-file /vscode/bin/linux
F1Help  F2Setup F3SearchF4FilterF5Tree  F6SortByF7Nice -F8Nice +F9Kill  F10Quit
```

htop (se instalado): mostra os processos do sistema em tempo real, com interface interativa que permite visualizar uso de CPU, memória e até encerrar processos diretamente.

5. Gerenciamento de Pacotes

```
● @AnaLaura1997 ➜~/aula_bigdata $ sudo apt install tree -y
  Reading package lists... Done
  Building dependency tree... Done
  Reading state information... Done
  tree is already the newest version (2.1.1-2ubuntu3.24.04.2).
  0 upgraded, 0 newly installed, 0 to remove and 131 not upgraded.
```

sudo apt install tree -y: instala o programa tree, que mostra a estrutura de pastas em formato de árvore.

```
● @AnaLaura1997 ➜~/aula_bigdata $ tree
  .
  └── arquivo1.txt

  1 directory, 1 file
```

tree :mostra a estrutura de diretórios e arquivos em formato de árvore, organizando visualmente as pastas e subpastas.

6. Permissões de Arquivos

```
● @AnaLaura1997 ➜~/aula_bigdata $ ls -l
  total 4
  -rw-r--r-- 1 codespace codespace 29 Mar  4 17:51 arquivo1.txt
```

ls -l: lista os arquivos e pastas do diretório atual mostrando detalhes como permissões, dono, tamanho e data de modificação.

```
@AnaLaura1997 ➜~/aula_bigdata $ chmod 755 arquivo1.txt
@AnaLaura1997 ➜~/aula_bigdata $ ls -l
total 4
-rwxr-xr-x 1 codespace codespace 29 Mar  4 17:51 arquivo1.txt
```

chmod 755 arquivo1.txt: Itera as permissões do arquivo para:

- 7 (dono) → pode ler, escrever e executar
- 5 (grupo) → pode ler e executar
- 5 (outros) → pode ler e executar

7. Compactação de Arquivos

```
@AnaLaura1997 ➜~ $ tar -czvf backup.tar.gz aula_bigdata/
aula_bigdata/
aula_bigdata/backup.tar.gz
aula_bigdata/arquivo1.txt
```

tar -czvf backup.tar.gz aula_bigdata/ : cria um arquivo compactado chamado backup.tar.gz contendo toda a pasta aula_bigdata e seus arquivos.

```
@AnaLaura1997 ➜~ $ tar -xzvf backup.tar.gz
aula_bigdata/
aula_bigdata/backup.tar.gz
aula_bigdata/arquivo1.txt
```

tar -xzvf backup.tar.gz: extrai o conteúdo do arquivo backup.tar.gz no diretório atual.

8. Teste de Rede

```
@AnaLaura1997 ➜~ $ sudo apt install iputils-ping -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  iputils-ping
0 upgraded, 1 newly installed, 0 to remove and 131 not upgraded.
Need to get 44.6 kB of archives.
After this operation, 124 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 iputils-ping amd64 3:20240117-1ubuntu0.1
44.6 kB]
Fetched 44.6 kB in 0s (127 kB/s)
Selecting previously unselected package iputils-ping.
(Reading database ... 58629 files and directories currently installed.)
Preparing to unpack .../iputils-ping_3%3a20240117-1ubuntu0.1_amd64.deb ...
Unpacking iputils-ping (3:20240117-1ubuntu0.1) ...
Setting up iputils-ping (3:20240117-1ubuntu0.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
@AnaLaura1997 ➜~ $ ping google.com
PING google.com (172.253.115.139) 56(84) bytes of data.
^C
--- google.com ping statistics ---
107 packets transmitted, 0 received, 100% packet loss, time 108548ms
```

ping google.com: testa se há conexão com o servidor do Google e mede o tempo de resposta da rede.

Dificuldade: precisei utilizar o sudo apt install iputils-ping -y antes para fazer a instalação.

```
● @AnaLaura1997 ➜~ $  curl ifconfig.me
○ 135.237.130.237@AnaLaura1997 ➜~ $
```

curl ifconfig.me: consulta um serviço externo e mostra o seu endereço IP público.

9. Instalação do Java (Pré-Hadoop)

```
● 135.237.130.237@AnaLsudo apt install openjdk-11-jdk -y
  Reading package lists... Done
  Building dependency tree... Done
  Reading state information... Done
  The following additional packages will be installed:
    alsa-topology-conf alsa-ucm-conf at-spi2-common at-spi2-core ca-certificates-java dbus-user-session
    dconf-gsettings-backend dconf-service fonts-dejavu-extra gsettings-desktop-schemas java-common
    libasound2-data libasound2t64 libatk-bridge2.0-0t64 libatk-wrapper-java libatk-wrapper-java-jni
    libatk1.0-0t64 libatspi2.0-0t64 libavahi-client3 libavahi-common-data libavahi-common3 libcups2t64
    libdconf1 libdrm-amdgpu1 libdrm-common libdrm-intel1 libdrm2 libgbm1 libgl1 libgl1-mesa-dri libglvnd0
    libglx-mesa0 libglx0 libice-dev libice6 liblcms2-2 libllvm20 libpciaccess0 libpcsclite1
    libsensors-config libsensors5 libsm-dev libsm6 libvulkan1 libwayland-client0 libx11-xcb1 libxaw7
    libxcb-dri3-0 libxcb-glx0 libxcb-present0 libxcb-randr0 libxcb-shape0 libxcb-sync1 libxcb-xfixes0
    libxcomposite1 libxi6 libxinerama1 libxkbfile1 libxmu6 libxmuu1 libxpm4 libxrandr2 libxshmfence1
    libxt-dev libxt6t64 libxtst6 libxv1 libxxf86dga1 libxxf86vm1 mesa-libgallium mesa-vulkan-drivers
    openjdk-11-jdk-headless openjdk-11-jre openjdk-11-jre-headless session-migration x11-utils
  Suggested packages:
    default-jre alsa-utils libasound2-plugins cups-common libice-doc liblcms2-utils pciutils pcscd
    lm-sensors libsm-doc libxt-doc openjdk-11-demo openjdk-11-source visualvm libnss-mdns
    fonts-ipafont-gothic fonts-ipafont-mincho fonts-wqy-microhei | fonts-wqy-zenhei fonts-indic mesa-util
```

sudo apt install openjdk-11-jdk -y : instala o Java Development Kit (JDK) versão 11, incluindo compilador e ferramentas para desenvolver e executar programas Java, sem pedir confirmação.

```
● @AnaLaura1997 ➜~ $ java -version
  openjdk version "25.0.1" 2025-10-21 LTS
  OpenJDK Runtime Environment Microsoft-12574222 (build 25.0.1+8-LTS)
  OpenJDK 64-Bit Server VM Microsoft-12574222 (build 25.0.1+8-LTS, mixed mode, sharing)
○ @AnaLaura1997 ➜~ $
```

java -version: mostra qual versão do Java está instalada no sistema.